

Matrix discrepancy and optimization

Global synchronization of a random cubic graph

Difficulty: challenging **Importance:** interesting to the community**Status:** Open**Provenance:** source-stated conjecture, with the phase domain written explicitly as a torus.**Rating rationale:** Controlling every local minimum on sparse random cubic graphs requires new landscape analysis beyond existing dense and high-degree results; it informs synchronization and nonconvex optimization.

For each even integer $n \geq 4$, let G_n be uniformly distributed over the labelled simple 3-regular graphs with vertex set $\{1, \dots, n\}$. For a graph G on these vertices, define

$$E_G(\theta) = \sum_{\{i,j\} \in E(G)} (1 - \cos(\theta_i - \theta_j)), \quad \theta \in (\mathbb{R}/2\pi\mathbb{Z})^n.$$

A phase vector is synchronized if $\theta_i = \theta_j$ modulo 2π for every pair i, j . A local minimum is understood with respect to the usual topology on the phase torus; it need not be strict.

Is it true that

$$\lim_{\substack{n \rightarrow \infty \\ n \text{ even}}} \mathbb{P}(\text{every local minimum of } E_{G_n} \text{ is synchronized}) = 1?$$

The energy is a sparse graph optimization problem whose derivatives involve weighted graph Laplacians. The conjecture asks whether a typical graph of the smallest plausible fixed regular degree has no nonsynchronized local minima.

References

1. A. S. Bandeira, A. Kireeva, A. Maillard, and A. Rödder, *Randomstrasse101: Open Problems of 2024*, arXiv:2504.20539 (2025), Definition 2.1 and Conjecture 3. [Paper](#).
2. P. Abdalla, A. S. Bandeira, M. Kassabov, V. Souza, S. H. Strogatz, and A. Townsend, *Expander graphs are globally synchronizing*, *Advances in Mathematics* 488 (2026), article 110773; arXiv:2210.12788v4, introduction and Section 7. The random regular result requires substantially larger fixed degree. [Paper](#).
3. V. Jain, C. Mizgerd, and M. Sawhney, *The random graph process is globally synchronizing*, arXiv:2501.12205 (2025), Theorem 1.2: the connectivity hitting-time result for a different random graph model. [Paper](#).
4. S. Lepsveridze and S. Zhang, *Graphs with connectivity $3/4 - \epsilon$ are globally synchronizing*, (2026), dense minimum-degree result. [Paper](#).

Status check — 2026-09-10

checked the current records 2504.20539v1 (2025-04-29), 2210.12788v4 (2026-01-19), 2501.12205v1 (2025-01-21), and 2608.20010v1 (2026-08-20), and searched “random cubic globally synchronizing 2026” and “random regular graphs degree 3 conjecture”. The large-degree, random-graph-process, and dense minimum-degree results do not establish the stated cubic-graph assertion. No proof, counterexample, or relevant withdrawal was found. The source’s angular variables give the torus used here; its isolated sphere notation is not imposed on the phases.

Audit update (2026-09-10): Rechecked the 2025 cubic-graph conjecture and August 2026 dense-degree theorem, and searched for later cubic synchronization results. The dense threshold result concerns a different graph regime and does not resolve this target. This is a bounded literature check, not a proof that no solution exists.